Factorial Stress-Testing of Claim Eligibility in Crop-Yield Models

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Abstract—Post-hoc explanations describe a fitted predictor, but the claims attached to them require separate predictive evidence. We stress-test a claim-eligibility audit across a pre-locked factorial crop-yield study with four weather representations, two hierarchy treatments, two search-grid sizes, five model families, and three feature families. Six nested rolling-origin outer folds produce 615 locked predictions after 34,560 inner jobs and 4,896 seeded outer fits. The selected stage-hybrid hierarchical CatBoost pipeline achieves pooled RMSE 0.461 t ha^{-1} and $R^2 = 0.110$, with positive R^2 in four of six folds. It passes overall predictive adequacy and incremental weather value, but fails the pre-specified event-recovery module. Thus, the evidence permits a population-level weather-specific predictive-reliance claim, not recovery of individual adverse events or causal attribution. A 28,800-row semi-synthetic suite reveals low permission power, while 600 PJM degradation runs show that the weather-value gate responds unevenly to different corruption mechanisms. Because selection and module estimation share the completed outer-fold leaderboard, the reported intervals are developmental rather than independent confirmation. The results position claim eligibility as a conservative scope controller, not a general validity certificate.

Index Terms—claim eligibility, crop yield, explainable machine learning, temporal validation, factorial experiment, reproducibility

I. Introduction

Feature attributions answer how a fitted prediction changes with its inputs; they do not establish that the prediction is accurate enough to support a substantive outcome claim. A coherent SHAP or LIME explanation can therefore be model-faithful while pointing in the wrong direction for the observed event [1], [2]. This distinction is especially consequential in crop-yield studies, where short panels, temporal drift, geographic structure, and correlated weather summaries make ordinary random validation optimistic [3], [4].

Claim-eligibility auditing addresses the prior question: what is the strongest interpretation licensed by locked predictive evidence? Module A requires a model to improve over a pre-specified naive comparator; Module B requires the feature group named in a claim to add predictive value; and Module E, invoked only for event-level claims, requires tail error, ranking, and top-k recovery. Passing a module narrows an interpretation boundary. It does not prove a causal effect, identify a data-generating mechanism, or eliminate leakage and shift [5], [6].

This paper extends an earlier fixed-design audit into a systematic factorial stress test. Its contributions are: (1) a fully crossed comparison of weather representation, hierarchical residualization, search-grid breadth, model family, and feature family under nested rolling validation; (2) a winner lock that ranks claim tier before predictive error and excludes all external-suite results; and (3) semi-synthetic and cross-domain calibration that reports both permission and abstention behavior. The central result is deliberately mixed: the selected crop pipeline passes Modules A and B but fails E. This supports a weather-specific predictive-reliance claim at population level, while blocking observed-event recovery and causal weather attribution.

II. Related Work and Claim Boundary

Machine-learning crop forecasts commonly combine weather, location, and trend information [7], [8]. Tree ensembles and gradient boosting can capture nonlinear interactions, but tuning and model choice must remain inside the training period [9]–[12]. Structured cross-validation is likewise necessary when observations share time or geography [4].

Model explanation and predictive permission are distinct. SHAP decomposes a fitted output; permutation or group ablation measures predictive reliance [1], [13]. Randomization tests show that visually plausible explanations can survive changes that should invalidate their interpretation [14]. We therefore treat explanations as model-descriptive by default. A stronger predictive claim is reported only when the corresponding locked gate passes. This resembles selective prediction at the study level rather than the row level [15].

The hierarchy used here is:

$$\begin{aligned}\text{description} &\subset \text{adequate prediction} \subset \text{group reliance}\\ &\subset \text{event recovery}.\end{aligned} \tag{1}$$

Each inclusion requires additional evidence. None implies causality.

III. Data and Target Construction

A. Crop panel and weather

The retrospective panel covers U.S. state-level barley, canola, oats, and wheat yields through 2025 from USDA NASS. Daily temperature, precipitation, and solar radiation come from NASA POWER [16], [17]. The analysis begins in 1990; each eligible crop–state series must provide at least three prior observations. At every inner or outer cutoff, a linear trend is fit using training years only. The target is the raw yield residual in t ha^{-1}, and event severity is scaled by the training-only residual standard deviation.

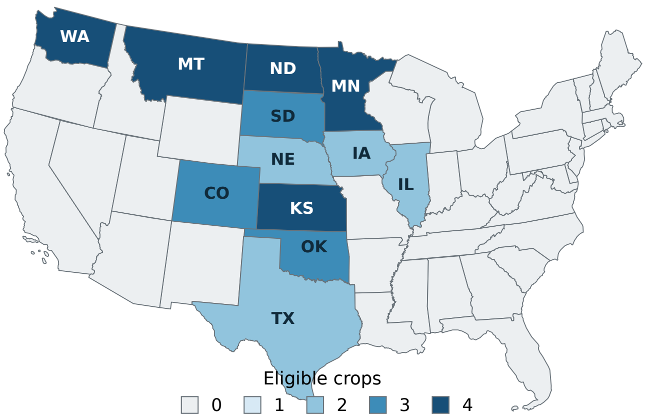

Fig. 1. Geographic coverage of eligible locked crop–state series. Shading reports the number of crops represented in each state; coverage is computed directly from the locked outer-fold row manifest.

B. Weather representations

BASE contains 35 full-season aggregates. STAGE-CAL and STAGE-GDD each contain 15 stage-specific summaries based on calendar progress or growing-degree-day boundaries. STAGE-HYBRID combines all three sources into 65 weather variables. Stage dates use the first USDA weekly report reaching 50% completion. Twelve cutoff-specific manifests preserve source and cutoff provenance.

Some early cutoffs did not contain three complete boundary-stage years. Before any model fitting, the protocol was amended to use medians from a frozen 1981–2025 USDA progress snapshot for those cells. Across the 12 manifests, calendar definitions comprise 120 frozen crop-season fallbacks, 88 frozen state fallbacks, 182 cutoff crop-season fallbacks, and 90 cutoff-safe state definitions. GDD definitions comprise 24 cutoff-safe progress definitions, 24 frozen crop-season fallbacks, and 12 fixed external canola fallbacks. No yield, validation, or model metric informed this amendment. It nevertheless introduces retrospective external-source information and is a limitation, not a claim of prospective deployability.

IV. Factorial Evaluation Protocol

A. Design and temporal separation

The design crosses four representations, FLAT versus HIER-RESIDUAL treatment, COMPACT versus EXPANDED hyperparameter grids, five estimators, and metadata-only, weather-only, and full feature families. HIER-RESIDUAL uses ridge partial pooling over crop, state, crop–state intercepts, and centered crop–state time slopes; unseen groups receive the population-level prediction. The model set is linear shrinkage, Extra Trees, LightGBM, CatBoost, and histogram gradient boosting. Table I summarizes the representation arms.

Six expanding outer folds test 2008–2011, 2012–2014, 2015–2017, 2018–2020, 2021–2023, and 2024–2025. Each outer training set contains three rolling inner folds. Hyperparameters minimize mean inner RMSE, with worst-fold

TABLE I
Weather representation component of the 16-arm design.

Representation	Arms	Weather vars.	Role
BASE	4	35 full-season	Reference
STAGE-CAL	4	15 calendar-stage	Fixed progress dates
STAGE-GDD	4	15 GDD-stage	Heat accumulation
STAGE-HYBRID	4	65 combined	BASE + CAL + GDD

RMSE and complexity as tie-breakers. Five fixed seeds quantify stochastic model variation. All preprocessing, trends, hierarchy terms, feature definitions, and selection are fit without outer-test outcomes.

The complete matrix contains 16 arms, 80 arm–model cells, and 240 feature-family pipelines. Completion was required before leaderboard creation: 34,560 of 34,560 inner jobs and 4,896 of 4,896 seeded outer fits finished. Across the complete leaderboard, 38/240 pipelines pass R1 and 122/240 pass Module A. Among 80 full pipelines, 46/80 pass B and 46/80 pass A+B; 0/46 E-evaluated candidates pass E.

B. Eligibility modules

For locked rows, define paired RMSE difference

$$\Delta(a, b) = \mathrm{RMSE}(a) - \mathrm{RMSE}(b). \tag{2}$$

Module A compares a candidate against the zero-residual baseline. Module B compares the full model against the same arm and estimator using metadata only. Each passes when the upper endpoint of a 2,000-draw calendar-year bootstrap 95% CI for Δ is below zero. A crop–state bootstrap is retained as a sensitivity check.

Module E is evaluated only after A and B pass. Event rows have training-scaled residual $z < -1$. E requires four conditions: upper 95% CI below zero for tail RMSE and MAE differences against zero, positive Spearman rank association with bootstrap lower endpoint above zero and permutation $p < 0.05$, and top-10 lift above one with bootstrap lower endpoint above one plus hypergeometric and within-year permutation $p < 0.05$. All conditions must pass.

C. Winner lock and external suites

Eligible pipelines first pass integrity and stable-performance rule R1: pooled $R^2 > 0$ and positive fold R^2 in at least four of six outer folds. Ranking then prioritizes claim tier (A+B+E, A+B, A, none), pooled RMSE, worst-fold RMSE, and complexity, with 0.001 RMSE tie tolerance. The winner and two runners-up were locked before either external suite was run. Because the winner was selected from the completed outer-fold leaderboard, its module intervals are post-selection developmental evidence rather than independent confirmatory intervals. Confirmation requires a prospectively locked new-year or external-region holdout.

EXT-SEMI perturbs the four locked candidates across five signal levels, four measurement-error levels, four spatial-mismatch levels, three event prevalence levels, and

TABLE II
Locked ranking after complete factorial evaluation.

Rank	Arm	Model	Family	RMSE	R^2	R^2>0	Tier
1	C310	catboost	full	0.461	0.110	4/6	A+B
2	C300	catboost	full	0.461	0.110	4/6	A+B
3	C301	extra_trees	full	0.462	0.106	4/6	A+B

Candidate-wide passes: R1 38/240; A 122/240; B 46/80 full; A+B 46/80 full; E 0/46 evaluated.

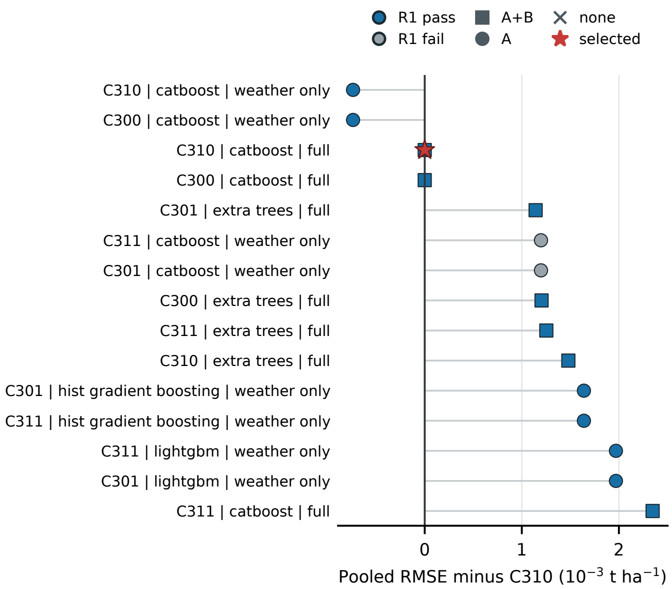

Fig. 2. Fifteen lowest-RMSE pipelines, shown as pooled RMSE difference from C310 on an expanded axis. Blue/gray denote R1 pass/fail; square, circle, and cross denote A+B, A, and none. The star marks the tier-first selection, which does not use external-suite results.

30 seeds. Its ground-truth labels score permission decisions and are never model or policy inputs. EXT-PJM uses hourly electricity demand from EIA-930 [18]; 600 seed-level runs degrade weather by permutation mix, noise, dropout, or lag substitution. It is cross-domain calibration only and cannot influence crop selection.

V. Results

A. Locked winner and fold stability

The winner is C310/CatBoost/full: STAGE-HYBRID, HIER-RESIDUAL, and COMPACT. It produces 615 pooled outer predictions with RMSE 0.461 t ha^{-1}, MAE 0.355 t ha^{-1}, and $R^2 = 0.110$. The selected pipeline has positive R^2 in 4/6 folds. C300 differs only by the FLAT treatment and is numerically tied within tolerance; C301 replaces CatBoost with Extra Trees and is slightly worse (Table II). The winner is determined by the locked hierarchy of rules, not by a claim that its point RMSE is uniquely superior.

Table III shows temporal heterogeneity. Performance is positive in O1–O4, nearly null in O5, and negative in O6. Pooled adequacy therefore should not be read as uniformly strong performance through time. The worst-fold RMSE is 0.610 t ha^{-1}, and the final fold R^2 is -0.218.

TABLE III
Outer-fold performance of locked C310/CatBoost/full.

Fold	Test years	RMSE	MAE	R^2
O1	2008–11	0.416	0.329	0.111
O2	2012–14	0.361	0.294	0.112
O3	2015–17	0.468	0.357	0.050
O4	2018–20	0.408	0.328	0.036
O5	2021–23	0.610	0.470	-0.002
O6	2024–25	0.485	0.366	-0.218
Pooled	2008–25	0.461	0.355	0.110

TABLE IV
Locked paired error statistics. Error differences are in t ha^{-1}; Module E uses $n = 136$ event rows.

Module	Comparison	Estimate	95% low	95% high
A	Full − zero	-0.028	-0.045	-0.013
B	Full − metadata	-0.028	-0.045	-0.013
E	Tail RMSE	-0.069	-0.102	-0.033
E	Tail MAE	-0.076	-0.110	-0.042

B. Claim-eligibility decision

Modules A and B both pass. Their paired RMSE difference is -0.028 t ha^{-1} with calendar-year bootstrap 95% CI [-0.045, -0.013]. The equality of A and B to displayed precision reflects the locked comparator predictions in this arm; it is not a shared test. Crop–state sensitivity upper endpoints are also negative (-0.0139 for each comparison).

Module E fails despite three favorable components. Among 136 event rows, tail RMSE difference is -0.069 t ha^{-1} (95% CI [-0.102, -0.033]) and tail MAE difference is -0.076 t ha^{-1} (95% CI [-0.110, -0.042]). Spearman $\rho = 0.287$ (bootstrap lower endpoint 0.123; permutation $p = 0.0141$). However, the predicted and observed top-10 sets overlap in only one row: lift 1.36, bootstrap lower endpoint 0, hypergeometric $p = 0.547$, and within-year permutation $p = 0.492$. Because E is conjunctive, top-k failure blocks event recovery. Table IV records the paired error components.

The resulting tier is A+B. It permits the statement that weather variables add predictive value to an overall adequate model in this retrospective population. It does not permit statements that the pipeline recovers particular low-yield events, that its feature attributions explain observed losses, or that weather caused those losses. The intervals quantify the selected developmental result; they do not independently confirm the winner after tier-first selection.

VI. External Calibration

A. Semi-synthetic permission power

EXT-SEMI contains 28,800 scored rows from 9,600 base fits. At base event prevalence, the locked winner's permission rate rises only from 4.0% at zero minimum detectable effect (MDE) to 8.8% at 1.5 MDE. The strongest runner rises from 9.8% to 20.0% (Table V). Figure 3 shows the complete signal curve by candidate.

TABLE V
EXT-SEMI A+B+E permission rate at base prevalence.

Candidate	0 MDE	1 MDE	1.5 MDE
winner	4.0%	6.5%	8.8%
runner_up_1	4.4%	6.0%	7.5%
runner_up_2	9.8%	14.4%	20.0%
kse_primary	3.8%	4.6%	5.0%

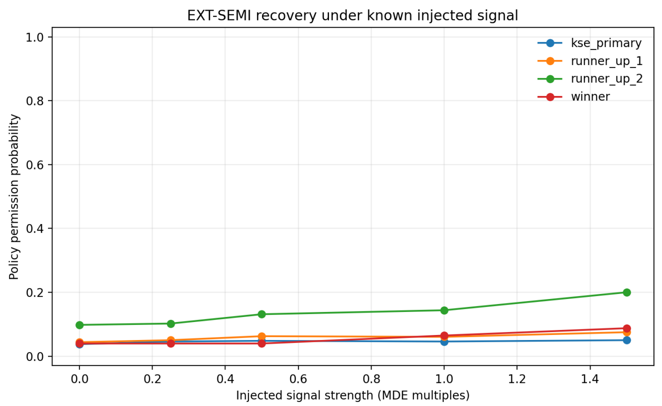

Fig. 3. Semi-synthetic permission probability versus injected signal at base event prevalence. Ground-truth labels are evaluation-only; decisions use observable Modules A, B, and E.

TABLE VI
EXT-PJM Module-B degradation endpoints.

Mechanism	Max. level	$P(B)$ at 0	$P(B)$ at max.
dropout	0.75	1.00	0.87
lag_substitution	7	1.00	0.00
noise	2	1.00	0.50
permutation_mix	1	1.00	0.83

These rates expose high false abstention under the tested data scale and policy. They should not be reframed as successful event sensitivity. The audit is conservative, and passing A+B in the crop panel does not resolve the low power of E.

B. PJM weather degradation

The PJM suite evaluates whether Module B responds when weather information is progressively degraded. After common train-median imputation, all $\lambda = 0$ inputs are invariant. At maximum degradation, B-pass probability is 0 for seven-day lag substitution, 0.50 for two-standard-deviation noise, 0.87 for 75% dropout, and 0.83 for complete permutation mixing (Table VI; Fig. 4). Thus, B is highly responsive to lag, partly responsive to additive noise, and comparatively tolerant of the tested dropout and permutation mechanisms.

This uneven response is informative: a gate that detects one kind of feature corruption need not detect another. Cross-domain evidence therefore calibrates behavior but does not validate agricultural transportability.

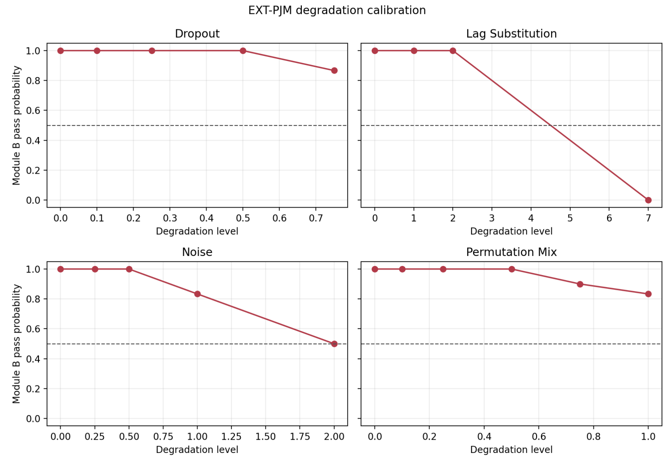

Fig. 4. Module-B pass probability across 600 PJM degradation runs. Curves are mechanism-specific calibrations, not crop-model selection evidence.

VII. Discussion

A. What the factorial study changes

The factorial result strengthens the developmental basis for an A+B claim in the evaluated retrospective population relative to a single hand-chosen pipeline. The locked winner emerges after comparing stage representations, hierarchy, grid breadth, models, and feature families under the same temporal protocol. Its nearest competitor is effectively tied, which argues for a conclusion about the attained claim tier rather than the uniqueness of C310.

At the same time, the experiment separates three questions that are often collapsed. Positive pooled skill answers whether useful prediction exists. Module B asks whether weather adds value beyond metadata. Module E asks whether the system can identify the specific adverse events that an explanation might be used to discuss. Here the first two answers are yes and the third is no. Feature-attribution outputs may therefore describe fitted weather reliance, but they cannot be presented as explanations of observed event losses.

B. Limitations

First, 615 locked rows are modest for a five-model factorial comparison, and two recent folds have non-positive R^2. Second, stage definitions partly use a frozen external crop-progress snapshot where cutoff histories are sparse. This was declared before fitting and uses no outcomes, but it remains retrospective information. Third, the semi-synthetic suite shows low A+B+E recovery even as signal grows; the policy can abstain from valid cases. Fourth, PJM is a useful mechanism check, not evidence that electricity-demand behavior transfers to crop yield. Finally, predictive reliance is not causal relevance. Unmeasured confounding, leakage, and geographic shift can preserve predictive contrasts [5], [6]. Moreover, the tier-first rule was fixed before execution but does not eliminate selection optimism when the same outer leaderboard selects and describes the winner. Independent confirmation requires a prospectively locked new-year or external-region holdout.

Future work should pre-register a prospective stage source, increase event counts, calibrate thresholds without weakening the event claim, and evaluate new years and held-out regions. Such studies should retain the rule that evaluation labels are never policy inputs.

VIII. Reproducibility

The protocol was locked after source-coverage audit and before model fitting. Checkpointed execution records all 34,560 inner jobs, 4,896 outer fits, 9,600 semi-synthetic base fits, and 600 PJM runs. A completion audit reports 30/30 items complete; the release QA reports 26/26 checks passed, including temporal access, row integrity, stage manifests, external-suite isolation, seed counts, and PJM zero-degradation invariance. The run manifest stores package versions, commands, file hashes, and the winner-lock hash. The frozen earlier manuscript and PDF are hash-checked and are not modified by this extension. An anonymized artifact containing the locked manifests, row-level predictions, completion audit, and figure-generation scripts is available at github . com / Anonymous21223 / Claim-Eligibility-Auditing/tree/ictai-v6.1-factorial.

IX. Conclusion

A full factorial search does not remove the need to limit explanation claims. The selected crop-yield pipeline passes overall adequacy and incremental weather value, but not event recovery. The defensible conclusion is therefore narrow: weather provides predictive reliance in the evaluated retrospective population, while explanations remain model-descriptive for individual adverse events. Semi-synthetic low power and mechanism-dependent PJM degradation further show that claim eligibility is a transparent scope controller, not a universal certificate of scientific validity. The result remains post-selection developmental evidence until tested on a prospectively locked holdout.

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